

TULA, Russian Federation

WMO: 277190

Lat: 54.2335N

Lon: 37.6135E

Elev: 205

StdP: 98.88

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.3	-21.0	-27.0	0.3	-24.1	-23.9	0.4	-20.4	8.7	-2.3	7.1	-1.5	1.0	10	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.2	20.0	28.2	19.2	26.5	18.5	21.2	27.6	20.3	26.4	19.4	25.0	2.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.2	24.1	18.2	13.4	23.1	17.3	12.7	22.3	62.2	27.8	59.2	26.4	56.2	25.2	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	DB	-28.5	32.0	3.3	2.4	-30.9	33.8	-32.9	35.2	-34.8	36.5	-37.2	38.3	(4)
(5)			WB	-28.7	22.7	3.2	1.7	-31.0	23.9	-32.9	24.9	-34.7	25.8	-37.1	27.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-7.5	-7.0	-1.8	6.7	13.6	17.1	19.7	17.5	12.0	5.9	-0.6	-5.1	(6)	
	DBStd	10.85	6.53	6.83	4.78	4.95	4.82	3.64	3.36	3.71	4.00	4.26	5.30	6.27	(7)	
	HDD10.0	2476	541	477	364	123	20	1	0	0	24	139	318	469	(8)	
	HDD18.3	4663	799	710	623	348	161	66	23	59	192	386	568	727	(9)	
	CDD10.0	1002	0	0	0	25	131	214	302	233	84	12	1	0	(10)	
	CDD18.3	147	0	0	0	0	13	30	67	34	3	0	0	0	(11)	
	CDH23.3	1358	0	0	0	4	140	263	588	339	23	0	0	0	(12)	
	CDH26.7	374	0	0	0	1	27	59	175	112	1	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.7	2.7	2.7	2.4	2.2	2.0	1.7	1.8	1.8	2.4	2.7	2.7	(14)	
(15) Precipitation	PrecAvg	625	43	37	34	42	51	71	76	65	62	56	47	43	(15)	
	PrecMax	788	94	80	81	85	87	179	153	238	170	110	112	96	(16)	
	PrecMin	374	15	10	12	16	14	22	19	10	7	14	4	17	(17)	
	PrecStd	95	18	15	18	18	21	36	34	52	39	28	24	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	4.0	13.1	23.5	29.5	30.6	33.5	34.1	25.9	19.6	12.3	7.5	(19)	
		MCWB	3.0	2.2	7.2	13.2	18.8	19.0	20.3	19.8	16.8	13.3	10.3	6.3	(20)	
	2%	DB	2.2	2.6	8.8	20.2	27.1	28.3	30.6	29.6	23.4	16.2	9.2	5.0	(21)	
		MCWB	1.4	1.3	4.3	11.5	16.8	19.0	20.1	20.3	16.2	12.5	7.7	4.2	(22)	
	5%	DB	1.3	1.7	6.1	17.7	24.6	26.5	28.5	26.8	20.9	13.7	7.5	2.9	(23)	
		MCWB	0.6	0.7	3.0	10.3	15.9	18.4	19.8	18.7	14.9	10.9	6.4	2.2	(24)	
	10%	DB	0.5	1.1	4.2	15.4	22.2	24.4	26.7	24.6	18.7	11.8	5.8	1.6	(25)	
		MCWB	-0.2	0.3	2.0	9.4	14.6	17.4	19.3	17.8	13.9	9.7	4.9	0.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.7	7.6	14.1	19.6	21.3	22.8	22.1	18.1	14.9	10.3	6.6	(27)	
		MCDB	3.7	3.5	13.2	21.4	27.8	27.7	29.1	28.6	23.6	17.5	12.1	7.3	(28)	
	2%	WB	1.6	1.6	5.0	12.5	17.9	20.2	21.5	20.7	16.9	13.0	8.0	4.3	(29)	
		MCDB	2.0	2.2	7.9	18.3	25.1	26.2	27.6	27.4	21.8	15.4	8.9	4.9	(30)	
	5%	WB	0.7	1.0	3.4	11.3	16.5	19.3	20.7	19.6	15.6	11.3	6.5	2.3	(31)	
		MCDB	1.2	1.5	5.4	16.6	22.7	24.7	26.7	25.8	19.6	13.4	7.3	2.8	(32)	
	10%	WB	-0.1	0.4	2.2	9.9	15.4	18.1	19.8	18.5	14.4	10.0	5.0	1.0	(33)	
		MCDB	0.5	1.1	3.9	14.6	21.1	22.9	25.2	23.6	18.0	11.6	5.8	1.4	(34)	
(35) Mean Daily Temperature Range	MDBR		5.3	6.2	7.0	9.4	10.6	9.9	10.1	10.2	9.1	6.5	4.4	4.5	(35)	
	5% DB	MCDBR	3.6	4.1	9.4	13.1	13.3	12.9	12.8	13.7	12.9	9.5	6.1	4.1	(36)	
		MCWBR	3.5	3.5	5.9	6.5	5.5	5.5	4.8	5.5	6.4	6.1	5.1	3.9	(37)	
	5% WB	MCDBR	3.5	3.9	8.0	11.6	12.1	11.3	11.2	12.0	11.0	8.1	5.4	3.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.332	0.350	0.422	0.392	0.383	0.422	0.434	0.387	0.357	0.327	0.293	(40)	
	taud		2.118	2.048	2.095	2.105	2.302	2.344	2.259	2.212	2.320	2.338	2.234	2.131	(41)	
	Ebn at Noon		634	723	810	798	854	864	819	781	773	708	590	542	(42)	
	Edh at Noon		67	102	121	139	121	117	125	124	98	77	61	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.74	1.67	3.18	4.11	5.48	5.71	5.52	4.60	2.94	1.42	0.58	0.42	(44)	
	RadStd		0.14	0.20	0.37	0.34	0.46	0.57	0.52	0.48	0.33	0.23	0.11	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	-252	+114	N/A	(46)
(47)	Regional (11 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-146	-221	+113	+38	(47)

Nomenclature: See separate page